

Peter Juhasz — Researcher

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Summary

- PhD Candidate in Mathematics
- Completed research projects in spatial statistics, network science, automated driving and complex systems
- Published multiple research papers and patented a Gaussian regression based trajectory prediction model

Education

- **PhD in Mathematics**
Aarhus University, Denmark, Stochastic Point Processes and Topological Data Analysis 2022–2025
Thesis: Point Process Based Models of Evolving Higher-Order Networks
- **Master of Business Administration with Highest Honors**
Budapest University of Technology and Economics, Hungary, Specialized in Finance 2019–2021
Thesis: Prediction of Popularity of Memes using Machine Learning Methods
- **Master of Science in Physics**
Budapest University of Technology and Economics, Hungary, Specialized in Applied Physics 2013–2016
Thesis: Examination of Spatially Embedded Complex Networks
Erasmus+ Scholarship: Delft University of Technology (Applied Physics)
- **Bachelor of Science in Physics**
Budapest University of Technology and Economics, Hungary, Specialized in Applied Physics 2010–2013
Thesis: Finite Element Method Plasma Simulation of Nitrogen Contaminated Ceramic Metal Halide Lamps

Professional Experience

- **Research Assistant in Spatial Statistics and Topological Data Analysis** 2022–
Department of Mathematics, Aarhus University, Denmark
 - Proved theorems in the field of stochastic point processes
 - Developed models for higher-order networks
 - Implemented topological data analysis algorithms
 - Taught 4 courses in mathematics and computer science
- **Deep Learning Researcher in Automated Driving** 2021–2022
Department of Automated Driving, Bosch Group Hungary
 - Patented a trajectory prediction model for automated driving
 - Real time object tracking with temporal convolutional neural networks
 - Object detection and classification with ultrasonic sensors
- **Software Architect in Automated Driving** 2018–2021
Department of Automated Driving, Bosch Group Hungary
 - Lead a team of 11 in environment prediction as a technical lead
 - Created a concept for automated ground-truth generation for data-driven machine learning development
 - Coordinated 2 teams in the integration of driver monitoring camera for hands-free driving
- **Software Engineer for Mobile Networks** 2016–2018
Business Unit IT & Cloud Products, Ericsson Telecommunications Hungary
 - Analyzed response times of distributed database servers for optimal load balancing
 - Coordinated a team of 8 as a deputy of the product owner
 - Implemented performance critical algorithms for Smart Services Routers

Publications, Patents

- Morten Brun, Christian Hirsch, Peter Juhasz, Moritz Otto
Random Connection Hypergraphs 2024
arXiv preprint, arXiv:2407.16334, 2024.
- Kate Barnes, Peter Juhasz, Marcell Nagy, Roland Molontay
Topicality boosts popularity: a comparative analysis of NYT articles and Reddit memes 2024
Social Network Analysis and Mining, 2024.
- Christian Hirsch, Benedikt Jahnel, Sanjoy Kumar Jhavar, Peter Juhasz
Poisson approximation of fixed-degree nodes in weighted random connection models 2023
arXiv preprint, arXiv:2311.12643, 2023.
- Christian Hirsch, Peter Juhasz
On the topology of higher-order age-dependent random connection models 2023
arXiv preprint, arXiv:2309.11407, 2023.
- Peter Juhasz
Information propagation in stochastic networks 2021
Physica A: Statistical Mechanics and its Applications, 577:126070, 2021.
- Peter Juhasz
Method and device for predicting the trajectory of a traffic participant, and sensor system 2020
CNIPA: CN113988353A, EPO: EP3916697A1, JPO: JP2021190119A, USPTO: US2021366274A1, 2020.
- Peter Juhasz, Jozsef Steger, Daniel Kondor, and Gabor Vattay
A Bayesian approach to identify Bitcoin users 2018
Public Library of Science One, 13:1–21, 2018
- Peter Juhasz, Szabolcs Beleznai, and Istvan Maros
Finite element method plasma simulation of nitrogen contaminated metal halide lamps 2014
Comsol Conference Proceedings, 2014

Awards, Grants

- **Academic Mobility Grant for International Exchange** 2024
Evolving Stochastic Network Models
20 000 DKK
- **Danish Data Science Academy PhD Fellowship (DDSA-PhD-2022-008)** 2022–2025
Topological Data Analysis Based Models of Evolving Higher-Order Networks
1 890 000 DKK
- **Hungarian National Bank Excellence Award for Outstanding Thesis and Diploma Work** 2021
Digitalization, Artificial Intelligence, and the Age of Data – What Makes a Meme Viral?
350 000 HUF

Programming Skills

- Python ● ● ● ● ●
- Git ● ● ● ● ●
- C++ ● ● ● ● ●

Languages

- Hungarian ● ● ● ● ●
- English ● ● ● ● ●
- German ● ● ● ● ●

Conferences, Talks

- **Poster: Simplex Count Distribution in Random Connection Hypergraphs**
Particle Systems in Random Environments 2024
- **Poster: Distribution of Betti Numbers in Age-Dependent Random Connection Models**
DDSA Annual Conference 2024
- **Talk: Topological Data Analysis Based Models of Evolving Higher-Order Networks**
Qualifying Exam 2024
- **Poster: Topological Data Analysis of Higher-Order Networks**
Danish-Swedish Summer School on TDA and Spatial Statistics 2023
- **Talk: Adaptive Trajectory Prediction Based on a Gaussian Regression Model**
Bosch Group: Bosch Innovation Day 2020
- **Talk: Probabilistic Localization of Bitcoin Users**
Hungarian Academy of Sciences: Statistical Physics Conference 2016
- **Poster: Dissecting Distributed Database Systems For Telecom Applications**
Ericsson Telecommunications Hungary: Ericsson University Day 2016
- **Poster: Finite Element Method Plasma Simulation of Ceramic Metal Halide Lamps**
Comsol Multiphysics Conference 2014

Teaching Experience

- **Data Project**
Aarhus University, Course Code: F24.550201U004.A 2024
10 ECTS Computer Project Class, Spring Semester
- **Linear Transformations**
Aarhus University, Course Code: E23.550171U010.A 2023
10 ECTS Exercise Class, Autumn Semester
- **Ordinary Differential Equations**
Aarhus University, Course Code: F23.550141U011.A 2023
5 ECTS Exercise Class, Spring Semester
- **Advanced Calculus for Engineers**
Aarhus University, Course Code: F23.280191U021.A 2023
10 ECTS Exercise Class, Spring Semester